AMARNATH CHAKRABORTY

GitHub • LinkedIn • Google Scholar

amarnathchakraborty95@gmail.com • achakraborty@mail.missouri.edu

BIO

Research Assistant, Department of Physics & Astronomy, University of Missouri–Columbia. My work centers on theoretical and computational modelling of two-dimensional quantum materials, with emphasis on transport, magnetism, and the impact of disorder on electronic states. I routinely employ tight-binding simulations, large-scale numerical diagonalization to investigate how external perturbations and lattice disorder reshape material properties—expertise directly relevant to two-dimensional semimetals, wide-bandgap systems, and semiconductor heterostructures. I am also exploring the integration of machine learning and quantum-simulation techniques into materials modelling. Looking ahead, I aim to develop modular hybrid workflows—linking physics-based simulation, AI-driven prediction, and high-performance computation—to accelerate the pathway from theoretical models to engineered materials and device-level functionality in real-world applications.

PI: Prof. Giovanni Vignale; Co-PI: Dr. Guang Bian.

EDUCATION

Ph.D. in Physics *2019–Present*
University of Missouri
Relevant Coursework: Quantum Mechanics 1 & 2, Electromagnetism, Solid-State Physics, Physics of Electronic Devices. Quantum computation through the course offered by IBM-QIC.

M.S. in Physics *2016–2018*
University of Hyderabad *Relevant Coursework*: Advanced Condensed Matter Physics, Theory of Quantum Materials, Numerical Methods.
CGPA: 8.18/10

PROFESSIONAL HIGHLIGHTS

- Developed and maintained Python/Julia simulation libraries for tight-binding and continuum models of twisted bilayer graphene; managed Git, CI, and documentation for cross-group reuse.
- Led three peer-reviewed research projects and coordinated closely with experimental collaborators (NUS).
- Awarded a competitive IBM quantum internship to implement and benchmark VQE/SQD workflows for the Fermi-Hubbard model.

TECHNICAL SKILLS

Programming	Python, Julia, Fortran, Qiskit, MATLAB, Mathematica
Quantum Computing	Quantum simulation in Qiskit (VQE, SQD), noise mitigation
Materials Modeling	Tight-binding, Kubo formalism, DFT (VASP), Green's function techniques
Mathematical Methods	Linear algebra, spectral methods, differential equations
Conceptual Areas	Topological phases, quantum algorithm & error correction, correlated electrons

RESEARCH INTERESTS

Bridging materials theory, ML, and quantum computing to accelerate discovery:

- Quantum geometry in Dirac/Weyl systems for targeted materials design.
- Lattice Hamiltonians for quantum simulation and resilient qubits.
- Variational and signal-processing quantum algorithms for correlated phases.
- ML-driven surrogates for 2D materials and high-T_c superconductors.

- Many-body methods (TMDCA, Lanczos) with active-learning for rapid screening.

RESEARCH EXPERIENCE

Doctoral Research, University of Missouri-Columbia (2019-Present) Focused on modeling and simulation of two-dimensional quantum materials for quantum technologies:

- **Numerical Modeling if 2D Quantum Materials (2024-Present)** Analyzed Berry curvature and orbital response by building continuum model of Twisted Bilayer graphene, as a potential candidate for designing of topological qubit platforms.
- **Variational Quantum Eigensolver Internship (Summer 2025)** Developing error-mitigated VQE and Sample-based Quantum Diagonalization for the Fermi-Hubbard model on IBM heavy-hex hardware, benchmarking correlated phase metrics.

Key Publications

- Insulator-Metal Transition & Magnetic Crossover in Bilayer Graphene.
  Phys. Rev. B 111, 125130 (2025)
- Giant Faraday Rotation in Nonsymmorphic Dirac Semimetals.
  Phys. Rev. B 107, 245120 (2023)
- Photon Absorption in 2D Nonsymmorphic Dirac Semimetals.
  Phys. Rev. B 105, 085101 (2021)
- Observation of 2D Weyl Fermions in Bismuthene.
  Nat Commun 15, 6001 (2024)

Master's Research, University of Hyderabad (2016-2018) Calculated polaron self-energies and decoherence factors in polar quantum dots via LLP Holstein methods—relevant to solid-state qubit design.

RESEARCH PROJECTS & ACADEMIC ENGAGEMENTS

- **Summer Schools**
  - **Qiskit Global Summer School**, IBM Q (2025) Hands-on workshops in quantum compilation, error mitigation, and VQE. Earned Quantum Excellence badge.
  - **Quantum Simulation Summer School**, UMD/NSF RQS (2025) Intensive training on near-term quantum simulation and resource estimation.
- **Internships**
  - **Quantum Research Intern**, Quantum Innovation Center, UMC (2025) Mentor: Dr. Pontus Laurell. Developing error-mitigated VQE and SQD for Fermi–Hubbard on IBM heavy-hex, benchmarking ground-state energies in ladder models.
  - **Summer Intern**, IIT Kharagpur (2017) Supervisor: Prof. S. S. Mandal. Solved the Weyl Hamiltonian via Boltzmann transport to analyze conductivity anomalies.
- **Visiting Research**
  - **Research Scholar**, I-FIM, NUS (2024) Studied orbital susceptibility and metal–insulator transitions in bilayer graphene under crossed fields.
- **Project Assistance**
  - **Project Assistant**, JNCASR (2019) Supervisor: Prof. N. S. Vidhyadhiraja. Applied TMDCA and Lanczos to compute LDOS/DOS and phase diagrams for disordered Hubbard models.
  - **Project Student**, JNCASR (2018) Supervisor: Prof. S. Sastry. Reproduced and analyzed Bouchaud's Trap Model and Soft Glassy Rheology framework.

CONFERENCES

- **IBM Quantum Developer Conference 2025**
  Presentation: Materials Modeling of Future: SKQD Approach of Fermi-Hubbard Model to investigate High Tc Superconductivity.
- **International Conference on Electronic Properties of 2D Systems (EP2DS-26) / MSS-22**, 2025
  Presentation: Trigonal Warping-Enabled Low-Field Insulator-Metal Transition in Bilayer Graphene

- **APS March Meeting**, 2021–2025
  - *Photon Absorption in Nonsymmorphic 2D Dirac Semimetals* (2021)
  - *Giant Faraday Rotation in Nonsymmorphic 2D Dirac Semimetals* (2022-2023)
  - *Insulator-Metal Transition and Magnetic Crossover in Bilayer Graphene* (2025)
- **Summer School on Entropy, Information, and Order in Soft Matter**, ICTS, 2018
- **National Science Conference on Condensed Matter Physics**, University of Hyderabad, 2017

FELLOWSHIPS AND AWARDS

Awards

- **Quantum Computation** -
  1. Achieved the Quantum Excellence badge (Advanced level) for successfully completing all the QGSS 2025 summer school assignments.
  2. Accepted as the Qiskit Advocate based on my achievements and experiences on Quantum simulation.
- **Physics Leaders Award** – Presented for outstanding research presentation in the Graduate Studies category at the University of Missouri.

National Scholarships and Fellowships in INDIA

- **INSPIRE Scholarship for Higher Education (2013–2018)** – Awarded by the Department of Science and Technology (DST), Government of India, for academic excellence in high school and commitment to pursuing pure science at the undergraduate level.
- **National Eligibility Test (NET), 2018** – Qualified for the Junior Research Fellowship (JRF) with an All India Rank (AIR) of 145.

Notable Academic Achievements

- Ranked in the top 1% nationwide in INDIA in the National Graduate Physics Examination.

OUTREACH AND INTERESTS

Science Communication & Outreach

- Participating **Quantum Computing** course for physics students at UMC, introducing Qiskit and topological qubit design principles (2025–Present).
- Active follower to **Quantum Open Source Foundation** discussions on GitHub projects for lattice Hamiltonian simulation tools.

Technical Hobbies

- Built a Python library for simulating Berry curvature effects in TBG under magnetic fields (GitHub: TBG-Bands & Response).

Personal Interests

- Avid reader of quantum computing literature (e.g., Preskill's notes, Kitaev's topological qubit papers).